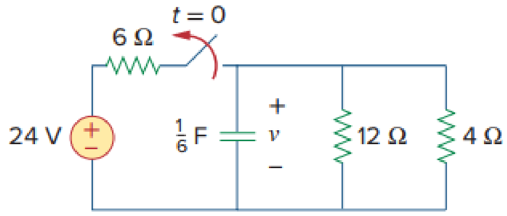
Chapter 7, Problem 2PP

Problem

If the switch in Fig. 7.10 opens at $t = 0$, find $v(t)$ for $t \geq 0$ and $wC(0)$.

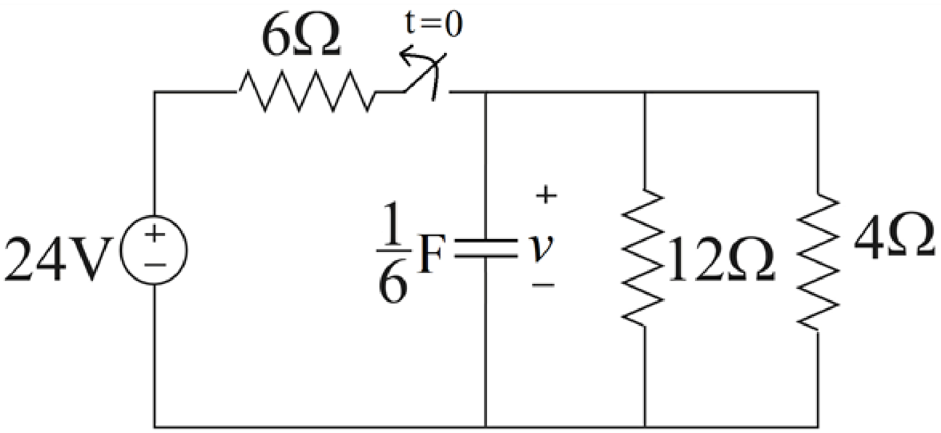
Fig. 7.10:



Step-by-step solution

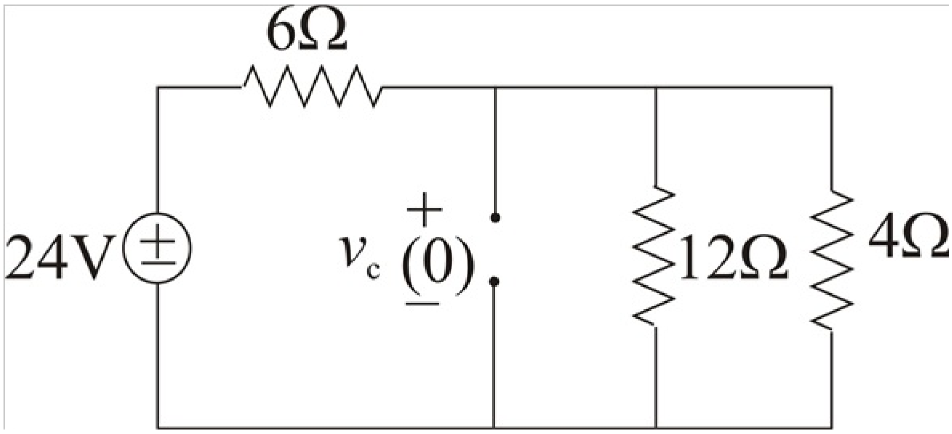
Step 1 of 9

Given circuit is



Step 2 of 9

For $t < 0$ switch is closed; the capacitor is open circuit to dc, and circuit is modified as



Step 3 of 9

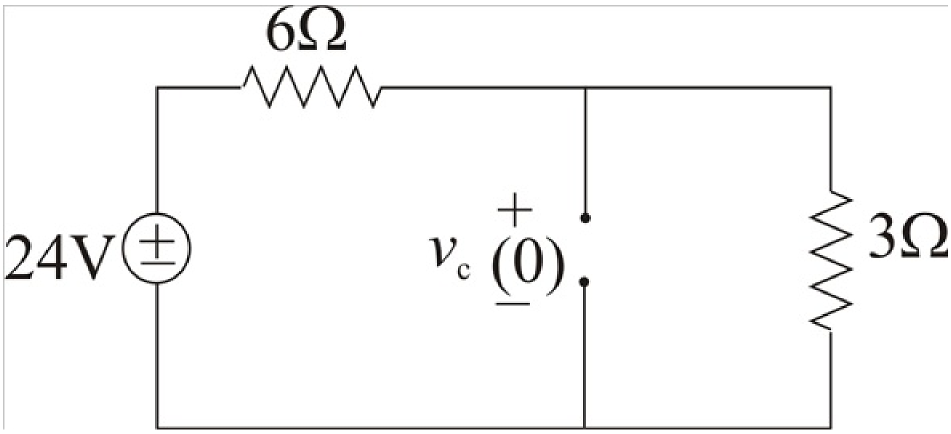
4Ω And 12Ω are in parallel

$$\therefore R_P = \frac{4\times 12}{4+12}$$

$$R_P = 3\Omega$$

Circuit modified as

Step 4 of 9



Step 5 of 9

By voltage division rule

$$v_{3\Omega} = 24\times\frac{3}{6+3}$$

$$v_{3\Omega} = 8\text{ V}$$

$$\therefore v_C(0) = 8\text{V}$$

The initial energy stored in the capacitor is

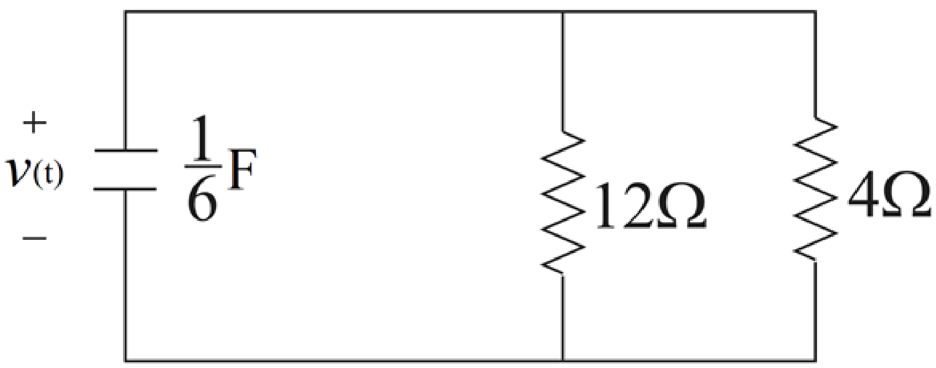
$$\omega_C(0) = \frac{1}{2}Cv_C^2(0)$$

$$\omega_C(0) = \frac{1}{2}\times\frac{1}{6}\times 8^2$$

$$\boxed{\omega_C(0) = 5.333\text{ J}}$$

For $t > 0$ switch is opened in the given circuit and equivalent circuit is as below

Step 6 of 9



Step 7 of 9

We need to find Thevenin's resistance across capacitor

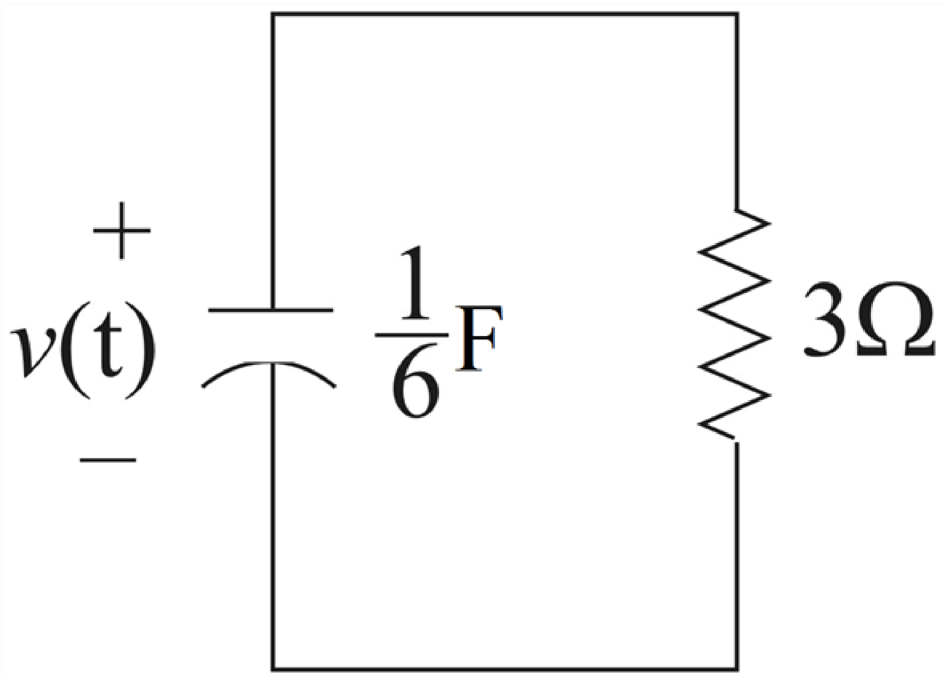
$$R_{Th} = 4 \| 12$$

$$R_{Th} = \frac{4 \times 12}{4+12}$$

$$R_{Th} = 3\Omega$$

Step 8 of 9

Above circuit becomes



Step 9 of 9

Time constant

$$\tau = R_{Th} C$$

$$\tau = 3 \times \frac{1}{6}$$

$$\tau = \frac{1}{2} \sec$$

Thus voltage across the capacitor for $t \geq 0$ is

$$v_C(t) = v_C(0) e^{-\frac{t}{\tau}}$$

$$\therefore v_C(t) = 8e^{-\frac{t}{\frac{1}{2}}}$$

$$\boxed{v_C(t) = 8e^{-2t} \mathrm{V}}$$